



Precursor solution chemistry *via* water additive enabling CZTSSe solar cells with over 12% efficiency

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ABSTRACT High-quality absorber films are imperative for highly efficient $\text{Cu}_2\text{ZnSn}(\text{S},\text{Se})_4$ (CZTSSe) thin-film solar cells. Precursor solution chemistry, interconnected with thin-film nucleation and the crystal growth process, should be studied insightfully. Thus, creative rationales and strategies could be proposed to improve the absorber quality, carrier characteristics, and hence device performance. However, the research on precursor solution chemistry in CZTSSe solar cells is still in its infancy. In this study, a facile eco-friendly additive strategy is reported to tailor the 2-methoxyethanol-based CZTS precursor solution chemistry. Incorporating water additives improves the homogeneity and thermogravimetric characteristics of a precursor solution by adjusting the particle sizes and coordination behavior inside the precursor solution. After the tailoring of the precursor solution chemistry, CZTSSe absorbers with high quality are fabricated. Hence, benefiting from the carrier dynamics enhancement, a CZTSSe device with a certified power conversion efficiency of 12.07% is achieved. The results open an avenue in precursor solution chemistry regulation through the use of eco-friendly additives to design highly efficient kesterite CZTSSe solar cells.

Keywords: CZTSSe solar cells, water additive, precursor solution chemistry, carrier characteristics

INTRODUCTION

To meet the increasing demands for the replacement of conventional fossil fuels, thin-film photovoltaic (PV) technology, particularly $\text{Cu}_2\text{ZnSn}(\text{S},\text{Se})_4$ (CZTSSe) PV, has shown impressive potential due to its high light absorption coefficient, earth-abundant and green constituents, adjustable bandgap, and flexibility [1–3]. Solution-based approaches demonstrate huge advantages over vacuum-based approaches due to the high material utilization (>95%) [4] and large-scale production potential with low cost. Solution-based methods are widely used in various highly efficient thin-film solar cells, such as $\text{Cu}(\text{In},\text{Ga})\text{Se}_2$ (CIGS) thin-film solar cells [5,6], rising perovskite thin-film solar cells [7,8], and CZTSSe thin-film solar cells [9,10]. In 2014, Mitzi's group [11] increased the power conversion efficiency (PCE) of CZTSSe solar cells to 12.6% based on the hydrazine solution method. In 2021, the recorded PCE of CZTSSe was improved to 13.0% by Xin's group [12], overcoming the stagnation in the PCE of CZTSSe solar cells over

seven years. Recently, Meng *et al.* [13] further improved the record efficiency to 13.6%. Not only are the overwhelming advantages of the hydrazine-based solution related to its precursor solution chemistry, but also the development of other solution-based methods has interconnection with the precursor solution chemistry.

The precursor solution chemistry regulates the CZTSSe absorber quality. As the main functional layer of CZTSSe thin-film solar cells, the quality of CZTSSe absorbers, including the crystallinity [14–16], defect properties, and carrier characteristics, determines the ultimate device performance. Therefore, an in-depth understanding of the precursor solution chemistry and effective regulation of the properties of a precursor solution are of primary importance. Guo *et al.* [17] found that it was difficult to realize metal-thiolate coordination in a thioglycolic acid-based precursor solution due to the difficulty in deprotonation for thiol groups to form R-S^- . Then, they introduced ammonia water in the precursor solution to tailor the deprotonation degree of thioglycolic acid and successfully regulated the coordination competition between the strong coordination (SH) and hydrophilic (COOH) groups. Benefiting from coordination engineering, they obtained CZTSSe films with good nucleation and crystallization properties, and thus a remarkable efficiency of up to 12.3% was achieved. In addition, the valence states of Cu and Sn were modulated to regulate the precursor solution chemistry, which affects the grain growth during selenization/sulfurization. Pan's group [18] utilized the high migration ability of Cu^+ (substituting CuCl_2 with CuCl for the precursor solution fabrication) to obtain a compact large-grain spanning (even extended over 5 μm) CZTSSe absorber. Gong *et al.* [19,20] found out that a precursor film fabricated from a Sn^{2+} -based precursor solution took a multi-step phase fusion reaction path during the selenization and exhibited SnSe_2 secondary phases on the surface. By contrast, a precursor film fabricated from a Sn^{4+} -based precursor solution went through a direct transformation reaction path and formed a uniform morphology. Furthermore, Ag [21], Cd [22], Ge [23], and alkali metal salts [2,24] were introduced into the precursor solution to optimize the precursor solution property and improve the crystallization of the CZTSSe absorber layer by researchers. Nevertheless, doping and alloying strategies increase the variety of elements, introduce more kinds of defects, and make the system highly complicated, exponentially increasing the difficulty of further improving the device performance.

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In this study, we first propose a simple water additive strategy for the 2-methoxyethanol (MOE)-based CZTS precursor solution. Environment-friendly water additives avoid the introduction of impurities, and this strategy fills research gaps in the precursor solution chemistry regulation in an MOE-based system. The competition between metal cations and Lewis base coordination interactions could be effectively regulated using water additives to obtain high-quality CZTSSe absorbers. A CZTSSe solar cell prepared by this strategy exhibits the highest PCE of 12.07% at the optimized amount of only 1 mol% (molar ratio to solvent) of water in the precursor solution.

EXPERIMENTAL SECTION

Materials

Copper acetate monohydrate ($\text{Cu}(\text{CH}_3\text{OO})_2 \cdot \text{H}_2\text{O}$, 99.95% metals basis), zinc chloride (ZnCl_2 , 99.95% metals basis), tin chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, 99.99% metals basis), MOE (99.7% GC), ammonia solution (NH_4OH , 25%–28%, AR), and cadmium sulfate 8/3-hydrate ($\text{CdSO}_4 \cdot 8/3\text{H}_2\text{O}$, 99.99% metals basis) were all purchased from Aladdin. Thiourea (TU, ACS 99.0%) was purchased from Sinopharm Chemical Reagent Co., Ltd. Selenium tablets (Se, 99.999%) were purchased from Huandao (Hebei) Technology Co., Ltd. All chemicals were used as received without further purification.

Fabrication of the precursor solution

ZnCl_2 (0.20 mol L^{-1}), $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (0.17 mol L^{-1}), $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{H}_2\text{O}$ (0.28 mol L^{-1}), TU (1.33 mol L^{-1}), and 8 mL MOE were sequentially added into the vial. After stirring at 60°C for 1 h, CZTS precursor solutions were successfully prepared. For the CZTS precursor solutions with different amounts of water additives, water was added dropwise in different molar ratios (1 mol%, denoted as 1%, and 2.5 mol%, denoted as 2.5%) to MOE after the addition of ZnCl_2 and $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ to facilitate their hydrolyzation. Then, $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{H}_2\text{O}$, TU, and MOE were added. After stirring at 60°C for 1 h, CZTS precursor solutions with different amounts of water additives were successfully prepared.

Device fabrication

Cleaned molybdenum-coated soda-lime glass substrates were used to fabricate CZTS precursor films. Approximately $1.5\text{-}\mu\text{m}$ -thick precursor films were fabricated after several times of “spin-coating (3000 r min^{-1} for 20 s) and baking (280°C for 120 s)” procedures. The procedure was performed under ambient air rather than in an inert atmosphere. Subsequently, precursor films were placed into a Se-containing graphite box for high-temperature selenization (550°C for 600 s) to fabricate CZTSSe absorbers. CdS buffer layers were deposited on CZTSSe absorbers through the chemical bath deposition method using a mixed solution of CdSO_4 solution, TU solution, and NH_4OH . Then, CZTSSe devices were fabricated with the typical architecture of soda lime glass (SLG)/Mo/CZTSSe/CdS/i-ZnO/indium tin oxide (ITO)/Ag/MgF₂. The CZTSSe devices were mechanically isolated with a full area of 0.2345 cm^2 .

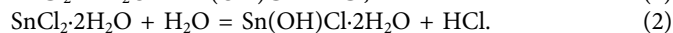
Characterization

Dynamic light scattering (DLS) measurements were carried out using Zetasizer Nano ZS90 (Malvern). Thermogravimetric analysis (TGA) was performed on SDT 650 (thermal analyzer)

under an N_2 atmosphere. The Raman spectra of the precursor solutions were obtained on inVia Renishaw using 785- and 532-nm excitation lasers with an exposure time of 10 s. X-ray diffraction (XRD) patterns were measured using PANalytical Empyrean 2 with Cu K α as the radiation source. Top-view and cross-sectional scanning electron microscopy (SEM) images of CZTSSe absorbers were collected using JSM-7900F (JEOL) with an electron acceleration voltage of 5 kV. EDAX Octane X-ray energy spectrometer equipped on JSM-7900F was used to measure the elemental composition under an electron acceleration voltage of 15 kV. Electron-beam-induced current (EBIC) measurements were conducted using the EBIC detector (A73C), which was installed on a TESCAN (VEGA3) SEM. Current density-voltage (J - V) curves were obtained on a Keithley 2400 source meter under illumination ($\text{AM } 1.5\text{G } 100 \text{ mW cm}^{-2}$), which was provided by a xenon-lamp-based solar simulator (Enlitech, SS-X50). The external quantum efficiency (EQE) was measured through the Enlitech QE-R test system. Capacitance-voltage (C - V) profiles were collected by an inductance-capacitance-resistance (LCR) meter (Keysight, E4990A) in the dark under a direct current (DC) bias range of -1 to 0.6 V by applying an alternating current (AC) voltage of 50 mV . Electrochemical impedance spectroscopy (EIS) measurements were also performed using the LCR meter by applying an AC voltage of 50 mV and a frequency range of 20 Hz to 2 MHz .

RESULTS AND DISCUSSION

The timing of adding H_2O is the most critical content in this work. As illustrated in Fig. 1, H_2O is added after weighing ZnCl_2 and $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and before adding the MOE solvent. After the addition of H_2O , the key step of the precursor solution modification, i.e., the hydrolysis of ZnCl_2 and $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (expressed as Reactions (1) and (2)), is completed. If H_2O is added with or after the MOE solvent, H_2O will be rapidly miscible with MOE and cannot react with ZnCl_2 and $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$. Zhao *et al.* [25] added water additives into a prepared dimethylsulfoxide (DMSO)-based precursor solution, and the amount of water was much more than that used in the current study. Hence, the coordination regulation that occurred in the DMSO-based precursor solution at the microscopic level could not be effectively realized. Therefore, only in this way (Fig. 1) can water additives play their role in tailoring the precursor solution chemistry. Compared with ZnCl_2 and $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, one of the Cl atoms is substituted by hydroxy in their corresponding hydrolysate. The existence of hydroxy could form hydrogen bonds with MOE, assisting in the formation of Zn- and Sn-related complexes as depicted in Fig. 1.



To investigate the effect of the coordination environment variation on the solvodynamic size distribution in the precursor solution, we conducted DLS measurements. As shown in Fig. 2a–c, the colloid sizes of all precursor solutions are larger than 1 nm . Hence, the precursor solutions we prepared are not true solutions, but cluster-containing solutions. A true molecular solution is certainly free of particles [26]. The average colloid size increases from 5.80 to 7.00 nm with the introduction of 1% water additive, whereas it decreases from 7.00 to 5.29 nm when the amount of the H_2O increases to 2.5%. The average colloid size increment with the introduction of 1% water additive may be attributed to the bridge role of MOE molecules and hydrogen

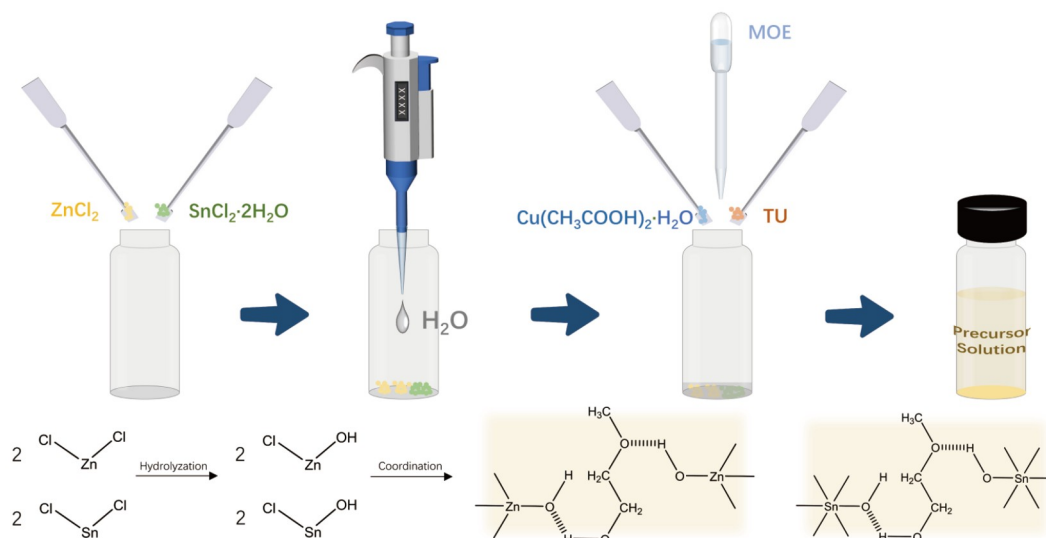


Figure 1 Schematic illustration of the fabrication of the water additive-containing precursor solution.

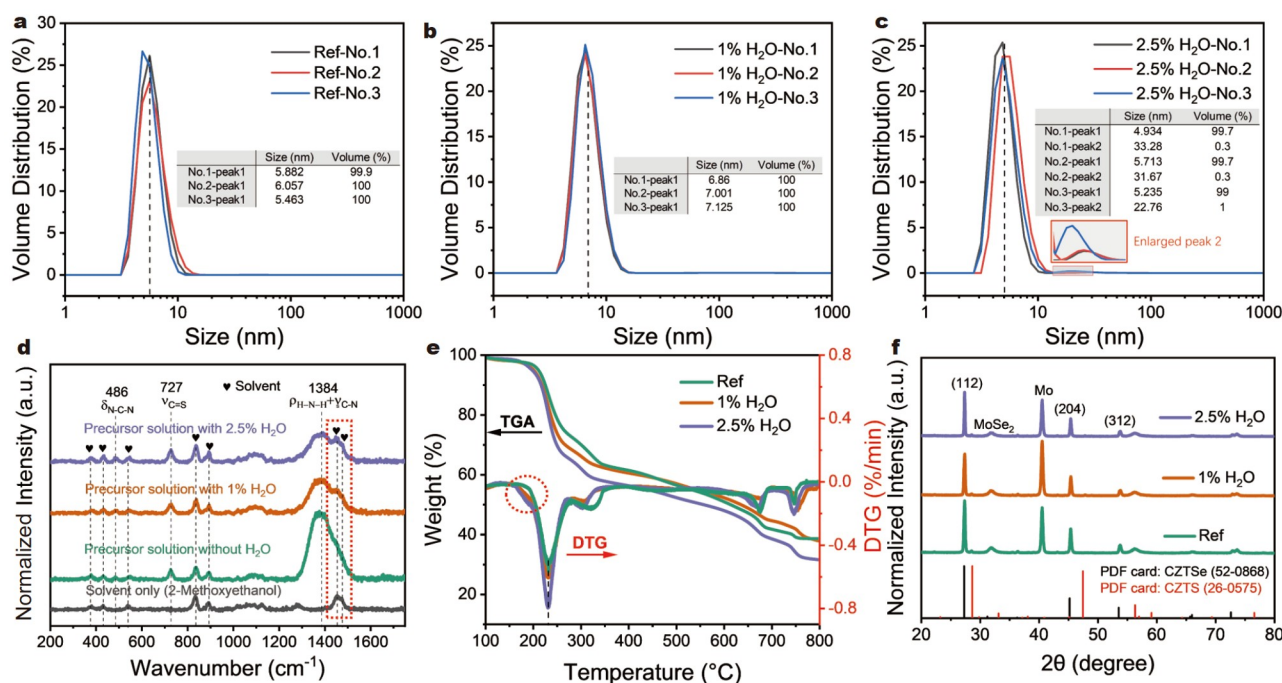


Figure 2 DLS spectra of the CZTS precursor solutions with (a) 0%, (b) 1%, and (c) 2.5% water additives. (d) Raman measurements and (e) TGA and DTG analyses of the precursor solutions with (a) 0%, (b) 1%, and (c) 2.5% water additives. (f) XRD patterns of the precursor films fabricated from precursor solutions with different amounts of water additives.

bonds between complex molecules after water addition, which promotes the formation of large-size complexes. With the introduction of 2.5% water additive, although the average colloid size is decreased, new peaks that also follow a Gaussian distribution centered in the range of 20 to 35 nm in DLS spectra are emerging. The larger cluster is probably a result of the hydrogen-bonding interactions between the MOE-metal complexes through excess H_2O molecules. The sizes of larger and smaller clusters in the three DLS measurement results are scattered, indicating that the precursor solution with excess water additive is not homogeneous. The best repeatability of the precursor

solution with 1% water additive implies its homogeneity and stability, which are some of the many advantages of the H_2O addition.

The vibrational modes of the solvated species were investigated by Raman spectra, as shown in Fig. 2d. The major active Raman modes for TU are the N-C-N bend vibration at 486 cm^{-1} , C-S stretch vibration at 727 cm^{-1} , and H-N-H rocking vibration and C-N symmetric stretching vibration at 1384 cm^{-1} [27,28]. Compared with the MOE solvent, TU-related vibration modes emerge, and MOE-related vibration modes at approximately 1450 and 1480 cm^{-1} (dotted rectangle in Fig. 2d)

disappear in the precursor solution without H₂O, i.e., reference precursor solution. With the increment of the amount of water, active Raman modes for MOE, especially the vibration modes at approximately 1450 and 1480 cm⁻¹, gradually increase, whereas the intensity of the TU-related vibration modes at approximately 1384 cm⁻¹ gradually decrease. The variation in the Raman spectra obviously manifests the difference in the coordination behavior between the precursor solutions with and without a water additive. This result is most likely due to the formation of the large-size complexes, which might change the coordination requirements of Zn and Sn atoms to MOE and TU molecules and make some vibration modes limited.

The TGA of the CZTS precursor with different amounts of water additives is shown in Fig. 2e. Differential thermogravimetry (DTG) curves are also presented to observe the weight-loss process and rate of the weight loss [29]. The first slow weight-loss up to about 190°C (dotted circle in Fig. 2e) is due to the evaporation of the residual MOE solvent. In the precursor with a water additive, an extra faster weight-loss stage in this temperature range makes the solvent evaporation a bit faster and more thorough, which may be attributed to the evaporation of water-MOE complexes. This result is consistent with the increasing intensity of MOE-related vibration modes at approximately 1450 and 1480 cm⁻¹ for the precursor solution with water additives in Fig. 2d. The major weight-loss stage was from approximately 190 to 230°C, mainly originating from the decomposition of TU and chlorides [30]. We speculate that the

partial decomposition of Zn(MOE)_x(TU)_y(OH)Cl or Sn(MOE)_x(TU)_y(OH)Cl complexes, as depicted in Fig. 1, also takes place in the major weight-loss stage. Thus, the evaporation of the MOE groups in the complexes generates more weight loss at this stage, whereas the content of impurities in the precursor films can be decreased. The following weight-loss stage from 230°C to approximately 320°C is mainly due to the decomposition of TU-metal complexes. The reference precursor owns a larger weight loss at this stage than the precursors with water additives, which may be due to the less TU participating in the coordination after adding water additives. Collectively, the water additive strategy can adjust the weight-loss process and decrease the content of impurities in the precursor films.

The different weight-loss process is bound to influence the morphology of the precursor films. In Fig. 3a, the reference precursor film has inhomogeneous cracks because of the incomplete evaporation of solvent molecules. Inhomogeneous precursor films lead to inhomogeneous absorbers with an uneven grain size. In the precursor film with 2.5% water, the precursor film is loose and consists of large nanocrystals (Fig. 3c), which may be due to its rapid and thorough weight-loss process during the baking. Moreover, large clusters detected in DLS in the 2.5% precursor solution may be the nucleation sites during the baking process and facilitate the growth of nanocrystals in the precursor films. Nevertheless, the excessively loose film is not beneficial for selenization. On the one hand, Se vapor is easily penetrated into the film and reacts with the Mo

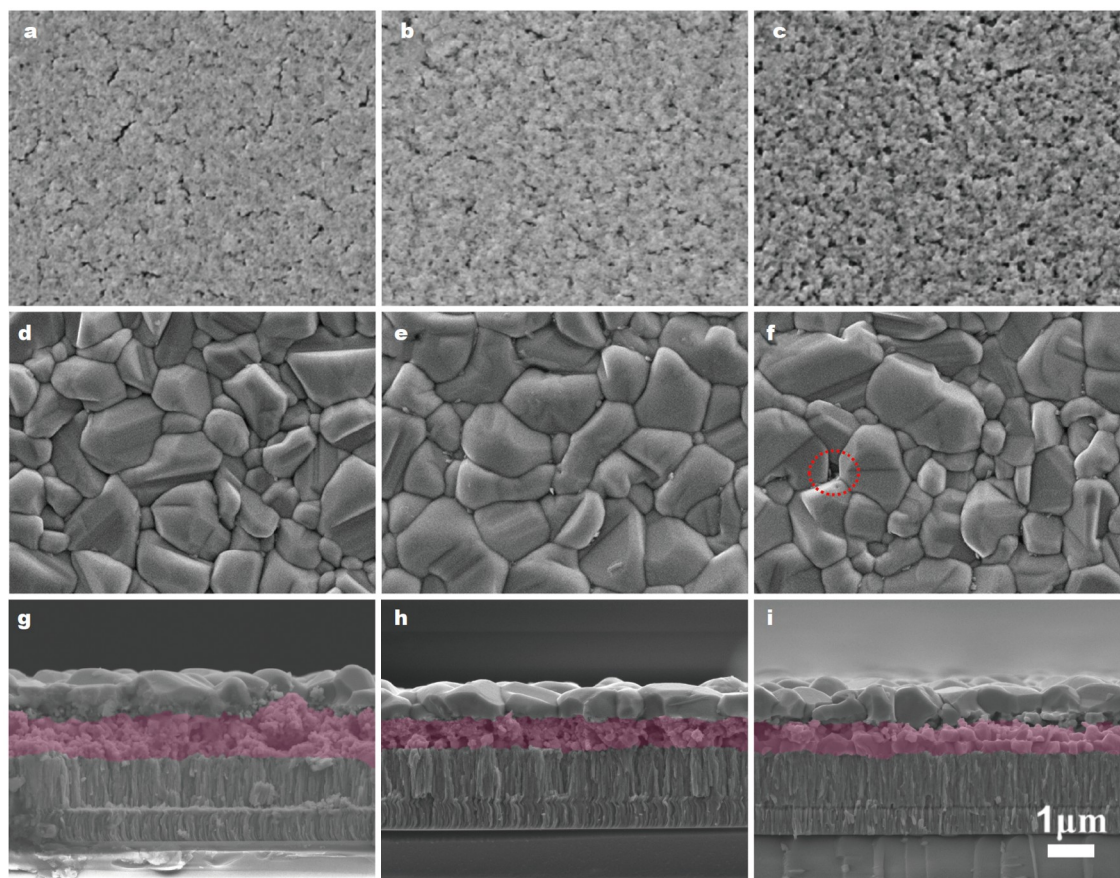


Figure 3 Top-view SEM micrographs of the precursor films with (a) 0%, (b) 1%, and (c) 2.5% water additives and absorbers with (d) 0%, (e) 1%, and (f) 2.5% water additives. Cross-sectional SEM micrographs of absorbers with (g) 0%, (h) 1%, and (i) 2.5% water additives.

layer to form a thick MoSe₂ layer, which will increase R_s [31,32]. On the other hand, the penetrated Se facilitates the formation of a Cu_{2-x}Se fluxing agent at the bottom of the film and hence a double-large grain layer absorber structure finally form (Fig. 3i) [16,33]. In the double-large grain layer structure, the grain growth at the bottom layer may take place at the expense of the grain quality at the top layer and hence result in voids at the surface in Fig. 3f. In addition, the bottom-large grain layer does not contribute at all to the current collection through EBIC measurements, as shown in Fig. S1. Instead, smaller grains in the middle have contributed to the current collection. This finding indicates that the bottom-large grain layer is useless, and the bottom layer of the absorber with 1% water, which consists of similar smaller grains, is beneficial for carrier collection. Comparatively, a precursor film with 1% water and the corresponding absorber both exhibit a homogeneous morphology, laying a good foundation for fabricating CZTSSe devices with good performance.

The XRD patterns of the selenized films are shown in Fig. 2f. The main diffraction peaks at 27.34°, 45.37°, and 53.79° correspond to the crystalline planes (112), (204), and (312) in the kesterite-type tetragonal phase, respectively. Combined with the Raman spectra (Fig. S2), no secondary phases exist in the selenized films fabricated from precursor solutions with different amounts of water additives. In addition, two peaks located at 31.93° and 56.29° for all the samples correspond to the (101) and (110) planes of MoSe₂, respectively. Obvious peaks imply a thick MoSe₂ layer (can also be seen in Fig. 3g–i) in the CZTSSe absorber, which suggests that there is room for further improvement. The structure, compactness, and crystalline plane orientation of the Mo layer all have an influence on the thickness of the MoSe₂ layer in our further research. The crystallite sizes extracted from the full width half maxima of the (112) plane

using the Scherrer formula for different selenized films are 36.3 nm for the reference film, 40.2 nm for the film with 1% water, and 77.7 nm for the film with 2.5% water. The crystallite size was increased with the increment of the amount of water, which may be attributed to the different nanocrystal sizes for the precursor films with different amounts of water additive. Moreover, as shown in Table S1, the water additive does not cause a composition fluctuation for CZTSSe absorbers.

To investigate the effect of water additives on the device performance, we measured the J - V curves of CZTSSe devices with different amounts of water additives under AM1.5 G illuminations, as illustrated in Fig. 4a. The device parameters are shown in Table 1. The statistic device parameters based on ten solar cells are box-plotted in Fig. S3. Using water additives has an influence on the open-circuit voltage (V_{oc}) and significantly affects the short-circuit current density (J_{sc}) and fill factor (FF). The J_{sc} and FF of the CZTSSe sample with 1% water additive are 36.22 mA cm⁻² and 67.46%, respectively, significantly higher than those of the reference CZTSSe sample (33.13 mA cm⁻² and 60.30%). However, all the other cell parameters were decreased by continuously increasing the amount of water additives, except for V_{oc} . The highest PCE of 12.07% (Fig. 4b) with a total area of 0.2402 cm² certified by the National PV Industry Measurement and Testing Center (NPVM) was obtained for the CZTSSe device with 1% water additive (Fig. S4). To explore the origin of the FF and J_{sc} improvement, the parameters of shunt conductance (G_{sh}), series resistance (R_s), diode ideality factor (A), and reverse saturation current density (J_0) of the solar cells were evaluated. In Fig. 4c, G_{sh} for each condition derived from the flat range is represented by a dashed line. According to the relationship between V_{oc} and parameters, including A , R_s , and J_0 , AkT/q is the slope, and R_s is the intercept of the linear fit to the data in Fig. 4d, whereas J_0 is the slope of the linear fit to the data

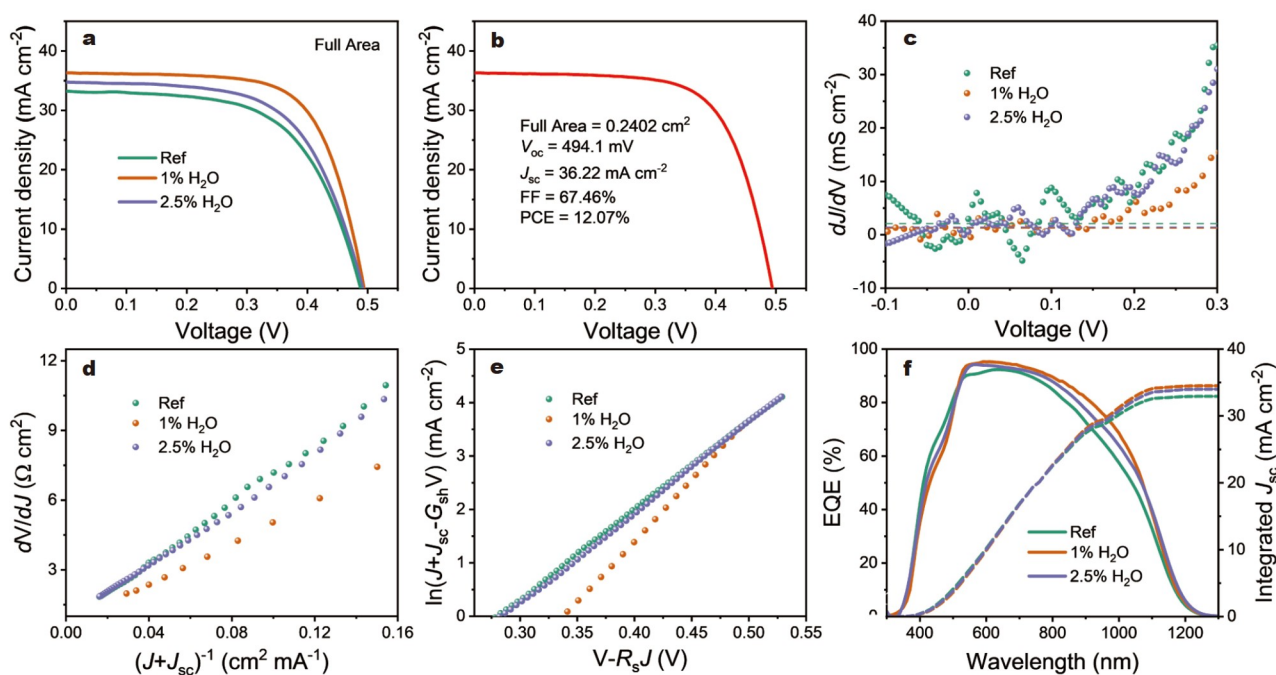


Figure 4 (a) J - V curves of CZTSSe devices with different amounts of water additive. (b) J - V curve of the certified cell with 12.07% efficiency. (c) dJ/dV versus voltage, (d) dV/dJ versus $(J + J_{sc})^{-1}$, (e) $(J + J_{sc} - G_{sh}V)$ versus $(V - R_sJ)$, and (f) EQE spectra of CZTSSe devices with different amounts of water additives.

in Fig. 4e. The detailed parameters are summarized in Table 1. The reduction in the diode factors, including G_{sh} , R_s , A , and J_0 , reveals CZTSSe absorbers with high quality and clearly indicates the much lower recombination rate of photogenerated carriers, producing an important enhancement in J_{sc} , FF, and hence PCE [34,35].

Based on the EQE measurement depicted in Fig. 4f, the integrated J_{sc} shows a similar trend to J_{sc} from the J - V curves. In addition, both devices with water additives show an improved spectral response at long wavelengths compared with the reference device. The enhancement in the spectral response is correlated with the improved generated charge-carrier collection, and less generated charge-carrier recombination in bulk and at the back interface of the absorbers [36]. The decreased spectral response from the CZTSSe sample with 1% to 2.5% water additive is attributed to the lower crystallization at 2.5% water additive. In addition, a sufficient selenization of the sample with 2.5% water additive makes the bandgap a bit smaller than that of the other samples (Fig. S5).

Fig. 5a shows the capacitance under a bias voltage from -1 to 0.6 V. The minority charge carrier density (N_{c-v}) and depletion width (W_d) can be extracted from the C - V profiles. The N_{c-v} versus W_d curves for the CZTSSe devices with different amounts

of water are plotted in Fig. 5b. The CZTSSe device with 2.5% water additive exhibited a lower N_{c-v} of $7.11 \times 10^{15} \text{ cm}^{-3}$ than that of the reference device and the device with 1% water additive ($1.22 \times 10^{16} \text{ cm}^{-3}$). Given that conventional C - V profiling consists of acceptors and trap density at the interface, low N_{c-v} of CZTSSe solar cells with water additives may be attributed to low acceptors or trap densities [37]. Relatively small bandgap while high V_{oc} for the CZTSSe devices with 2.5% water additive verifies that low trap densities may be the main cause of a low N_{c-v} . Therefore, water additives are beneficial to defect passivation in CZTSSe solar cells. Furthermore, W_d is gradually enlarged with the increment of the water additive amount. An enlarged W_d is helpful for the carrier collection at the heterojunction. However, an overlarge W_d is not beneficial for CZTSSe devices. A relatively low carrier lifetime and short diffusion length make partial carriers highly possible to recombine or decay before they are collected, and hence charge carriers could not be effectively collected when W_d is overlarge. The trade-off between defect densities and W_d is optimized at 1% water additive.

To further investigate the charge transport behavior of the devices, the built-in potential (V_{bi}) and impedance were analyzed. The V_{bi} estimated by the slope of $1/C^2$ versus the bias

Table 1 Detailed parameters of CZTSSe devices with different amounts of water additive

	J_{sc} (mA cm^{-2})	V_{oc} (V)	FF (%)	Efficiency (%)	G_{sh} (mS cm^{-2})	R_s (cm^{-2})	A	J_0 (mA cm^{-2})	W_d (nm)	N_{c-v} (cm^{-3})
Ref.	33.13	0.488	60.30	9.76	2.08	0.73	2.46	9.42×10^{-3}	137	1.32×10^{16}
1% H_2O	36.22	0.494	67.46	12.07	1.27	0.54	1.74	4.15×10^{-4}	159	1.22×10^{16}
2.5% H_2O	34.72	0.492	61.23	10.47	1.45	0.81	2.31	7.33×10^{-3}	176	7.11×10^{15}

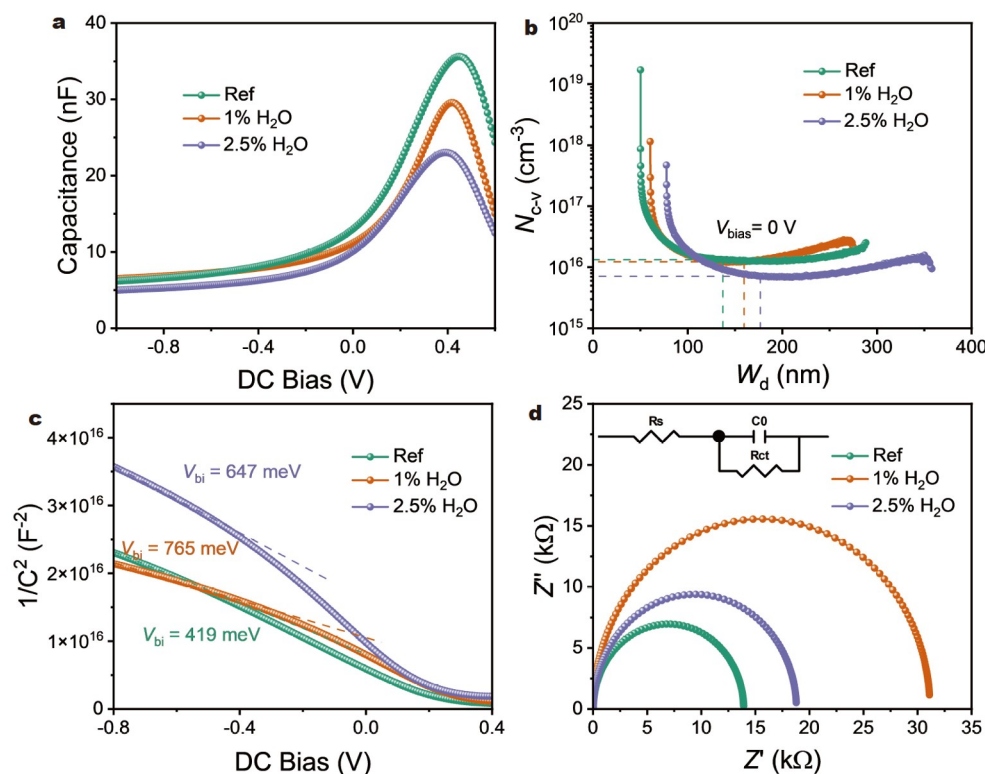


Figure 5 (a) C - V curves and (b) carrier concentration and depletion width of CZTSSe devices derived from the C - V curves of CZTSSe devices with different amounts of water additives. (c) Plots of C^{-2} versus DC bias of CZTSSe devices with different amounts of water additives to determine the built-in voltage. (d) Nyquist plots of CZTSSe devices with different amounts of water additives.

voltage is plotted in Fig. 5c. The highest V_{bi} of the CZTSSe device with 1% water additive among the three conditions results in the strongest electric field, which is favorable for enhancing the carrier separation and charge transfer, resulting in the most efficient device [38]. According to the EIS measurements, the effective lifetime of charge carriers can be obtained from the relation $\tau = R_{ct} \times C_0$ based on the equivalent circuit (a single parallel recombination R_{ct} and chemical capacitance C_0 connected to a series resistor R_s) depicted in the inset of Fig. 5d [39]. The prolonged carrier lifetime in the CZTSSe device with 1% water additive exhibits a significant improvement in the carrier transport dynamics. Furthermore, the decreases of R_s and increases of R_{sh} can be verified from the Nyquist plots.

CONCLUSIONS

In summary, a facile strategy of MOE-based CZTS precursor solution chemistry regulation was developed to fabricate high-efficiency CZTSSe solar cells. Water additives effectively regulated the coordination behavior in the precursor solution, resulting in a homogeneous and stable CZTS precursor solution. The optimized thermogravimetric characteristics of the precursor solution modulated the absorbers' morphology and crystallinity, laying a good foundation for device efficiency enhancement. Furthermore, the promoted carrier transport dynamics derived from a large depletion width, high V_{bi} , and long effective carrier lifetime resulted in a notable improvement in J_{sc} and FF. Finally, the CZTSSe solar cell manufactured using 1% water additive achieved a high certified PCE of 12.07%. Our results provide a new direction in exploring the precursor solution chemistry regulation *via* impurity-free and eco-friendly additives in CZTSSe solar cells.

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Author contributions Liu F supervised the whole project. Zhao X designed the experiment and drafted the manuscript. Zhao X, Pan Y and Liu S carried out the experiments for device fabrication, optimization, and characterization. Pan Y, Jiang L, and Lai Y discussed the results and provided valuable suggestions for the manuscript.

Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Supporting data are available in the online version of the paper.



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基于水添加剂策略调控前驱体溶液化学制备效率超过12%的CZTSSe太阳电池

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摘要 高质量的吸收层对制备高效率Cu₂ZnSn(S,Se)₄ (CZTSSe)薄膜太阳电池来说至关重要。与薄膜成核和晶体生长过程相关的前驱体溶液化学值得深入研究,从而有助于提出创造性的原理和策略来提高吸收层的质量和载流子传输特性,进而提升器件性能。然而,前驱体溶液化学的相关研究在CZTSSe太阳电池领域仍处于起步阶段。本工作报道了一种简便的环保添加剂策略来调控基于乙二醇甲醚体系的前驱体溶液化学。加入水添加剂可通过调整前驱体溶液内的溶胶粒径和配位行为来改善前驱体溶液的均一性和热重特性。成功对前驱体溶液化学进行调控后,得到了高质量的CZTSSe吸收层。在此基础上,得益于载流子动力学的增强,最终获得了认证效率达到12.07%的CZTSSe器件。本研究为通过环保添加剂调控前驱体溶液化学从而制备高效率锌黄锡矿CZTSSe薄膜太阳电池开辟了道路。